

Study Planner Tool

1. Project Idea and Justification

Project Idea

Study Planner Tool for university students.

Justification

Most students find assignments, exams and project deadlines difficult to track. Research indicates that students often forget about deadlines because they are not organized. Most of the tools today are complex and have features that the student does not need.

The goal of this project is to create a simple, uncomplicated study planner tool where students can keep track of all their tasks, deadlines and progress in one place. Additional features such as focus mode for students to focus on one task at a time, and extending deadlines when needed. It will also be easy to use for beginners.

2. Problem Statement

Normally students forget some of their tasks when they enroll in more than one course at the same time and they find it hard to keep track of each task for each course. Studies show that student's time management capabilities are weak and they are one of the main reasons for missed deadlines. It is easy to forget sometimes, for example you forget to turn in your homework or your project on time and that has an impact on the student's grade. Usually the student has a very hard time focusing on one task because they will think about other tasks that they will have to turn in soon. And that's why they need a way (a simple way) to track what they are working on and extend the deadlines if needed, and prioritize what really needs to be focused on in this particular moment so they don't get distracted.

3. Scope of Work

In-Scope (Core Features)

The system will have these features:

- Add/Edit/Delete Tasks (Assignment, Exam, Project).
- Add or extend task deadlines (+1/+3/+7/+Custom Days).
- View tasks in chronological order with high priority tags (Red text for High priority and Yellow text for Medium priority tasks).
- Use full-screen mode to view only the incomplete task with higher priority.
- Change task status to either Not started, Progress or completed.
- Sort tasks by deadline/priority/task type/status.
- Alert student is overdue or due today.
- View the percentage completed each week using a chart (Completed, Pending or Overdue).

User Roles

Student (Main User):

- Manage tasks.
- Track and extend deadlines.
- Update task status.
- Activate Focus Mode.
- View progress statistics.

Constraints

- The system will be simple (no complex backend).
- No internet connection (offline system).
- individual student use only (no collaboration).
- Focus Mode shows one task each time.

Verification Process

- Verify for adding and saving tasks.
- Verify if the deadlines show up and if they can be extended.
- Verify if Focus Mode displays only the highest priority task.
- Verify if overdue task generate notification.
- Verify if the progress stats truly represent task completion.

Out of Scope

- Recommendations based on AI or automatic scheduling.
- Online or cloud synchronization.
- Payment or subscriptions.
- Collaboration features.
- Collaboration with calendar apps.

4. Project Objectives

- Help students keep track of deadlines.
- Simple study planner application.
- Reduce deadline misses.
- Reduce distractions with focus mode.
- Easily extend a deadline with the extend deadline feature.
- Help students manage their tasks.
- Allow students to track their progress and stay motivated by seeing their achievements.

5. Project Timeline

Week	Task	Milestones	Completion Date
Week 1	Idea selection, planning	Proposal approved	14/10/1447
Week 2	Writing problem statement, scope	Planning completed	20/10/1447
Week 3	Requirements analysis	Requirements ready	6/11/1447
Week 4	User analysis, use cases	Analysis completed	13/11/1447
Week 5	UML diagrams	Design started	20/11/1447
Week 6	UI design	Design completed	27/11/1447
Week 7	Testing scenarios, cases	Testing completed	5/12/1447
Week 8	submission	Project submission	18/12/1447

6. Requirements Analysis

Functional Requirements

The system shall provide the following functionalities:

Task Management

- The system shall allow the user to add a task with: title, type (Assignment / Exam / Project / Lab / Reading), deadline (date & time), priority level (High / Medium / Low), estimated time, and optional notes.
- The system shall allow the user to edit any task.
- The system shall allow the user to delete a task with a confirmation dialog.
- The system shall save every change automatically after every operation.

Deadline Management

- The system shall sort tasks to nearest deadline by default.
- The system shall highlights overdue tasks in red and tasks due today in yellow.
- The system shall give the user the option to extend a deadline with quick options of +1 Day, +3 Days, +1 Week, or Custom date and time.
- The system shall provide an optional reason field when extending a deadline.

Task Status Tracking

- The system shall allow the user to change tasks status as: Not Started, In Progress, or Completed.
- The system shall separate completed tasks from active tasks.

Focus Mode

- The system shall have a button on the Dashboard that says “Focus Mode”.
- The system shall show only the highest priority task If Focus Mode is activated, and none of the other dashboard items will be visible when Focus Mode is on.
- The system shall show title, time remaining, and priority label in the task card.
- The system shall allow the user to Mark the task as Complete or Extend Deadline.

Dashboard & Overview

- The system shall show a summary of: Total Tasks, Overdue Tasks and Due Today.
- The system shall display a progress bar showing task percentage.
- The system shall early view the upcoming tasks on the Dashboard.

Filter & Sort

- The system shall allow filtering tasks by: type, priority, overdue, due today or due this week.
- The system shall allow sorting by: nearest deadline, priority, recently added, subject.

Notifications

- The system shall alert the user when a task is overdue.
- The system shall remind the user when a task is due within 24 hours.
- The system shall confirm when a task is marked as completed.

Progress & Stats

- The system shall show a weekly bar chart showing tasks completed per day.
- The system shall display a donut chart showing completed vs pending vs overdue tasks.
- The system shall show a motivational message based on weekly performance.

Non-Functional Requirements

- **Usability:** easy to learn for beginners to know how to operate with in 2 minutes (no training period).
- **Performance:** Response time ≤ 2 seconds for any operation (Fast) .
- **Reliability:** Chance of losing data should be less than 1% for normal operation.
- **Availability:** Available anytime and every day (No internet needed).
- **Maintainability:** The code is modular design (UI, logic, storage separate from eachother).
- **Security:** Data is stored locally on user device with limited access restriction.
- **Accessibility:** UI indicators contain text labels and not just colors (so colorblind users can use it too).

7. User Analysis

Target Users

- University students.
- School students.

User Personas

Persona 1: Sara

- **Age:** 20 Years Old.
- **Grade:** Second year in Computer Science.
- **problems:** Has more than one deadline coming up for different classes, can't focus when there is too much to do, forgets assignments.
- **Goals:** Keep track of every thing in one place, easy to see what I need to do soon, be able to focus on doing one thing at a time.
- **Behavioral Traits:** Uses mobile apps daily, only uses apps that have simple and clean UI, studies for short periods to stay focused.
- **System Requirements:** Focus Mode, sorts by deadline, highlights when something is past due, shows how long you are in a task.

Persona 2: Ahmed

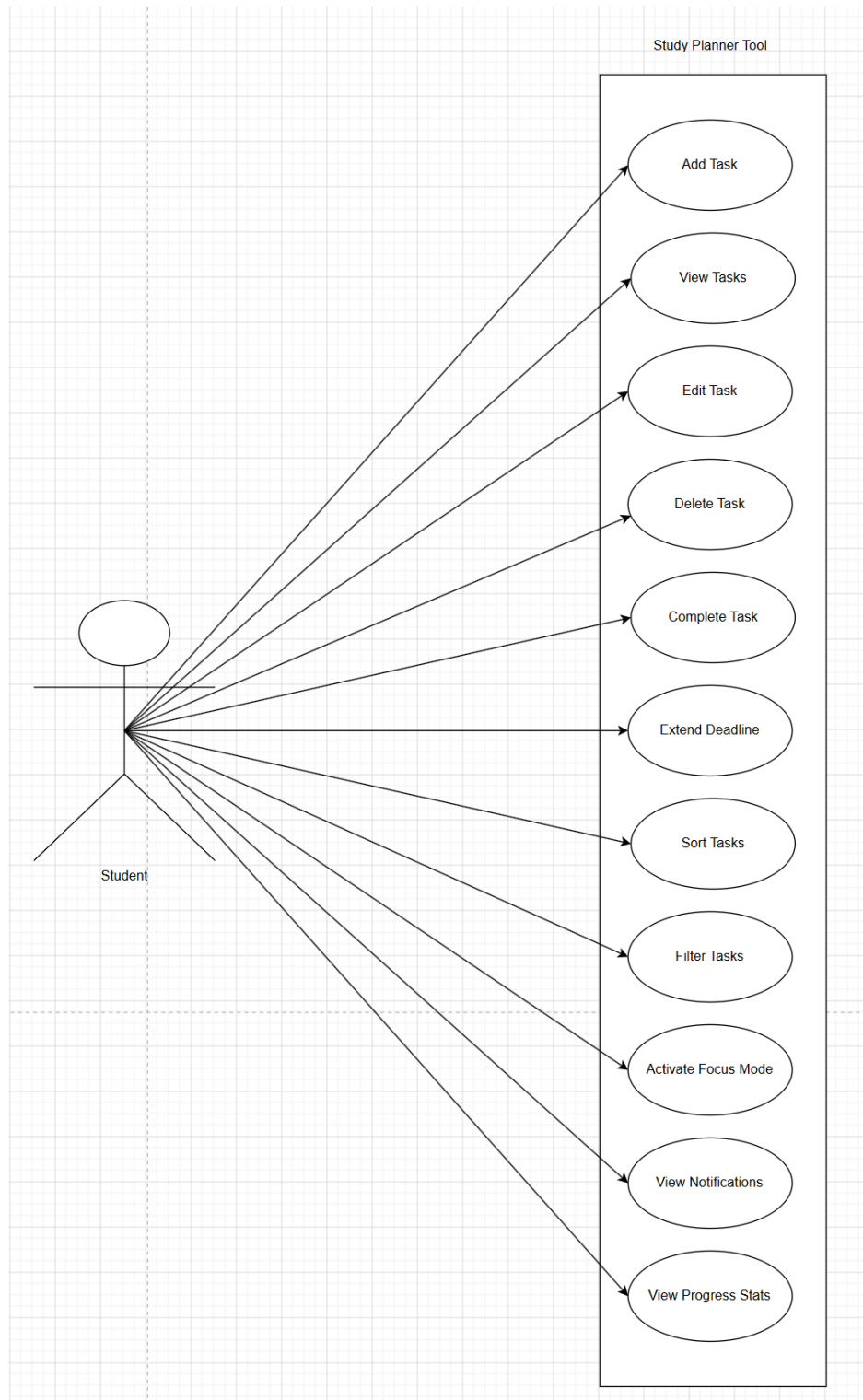
- **Age:** 18 Years Old.
- **Grade:** Highschool student.
- **problems:** Bad time-management skills, easily distracted, sometimes needs to extend deadlines.
- **Goals:** Quickly see what needs to be done, extend deadlines if needed without getting behind.
- **Behavioral Traits:** Short term apps user, avoids anything complicated, requires simple features.
- **System Requirements:** Show simple list of tasks, fast adding, ability to extend a deadline, notify me.

8. Use Cases

Primary Actor: Student

Use Case	Description
Add Task	User creates a new task with title, type, deadline, priority, and notes.
View Tasks	User sees all tasks sorted by nearest deadline.
Edit Task	User edits any task details in place.
Delete Task	User confirms and deletes a task.
Complete Task	User marks a task as completed.
Extend Deadline	User extends a task deadline using quick options or custom date.
Sort Tasks	User organizes tasks by deadline, priority, or type.
Filter Tasks	User filters tasks by type, priority, or status .
Activate Focus Mode	User activates Focus Mode, seeing only the most urgent task. User receives overdue alerts and reminders for upcoming deadlines.
View Notifications	User sees overdue alerts and upcoming deadline reminders .
View Progress Stats	User sees weekly completion chart and donut chart.

9. Use Cases Diagram



10. Risk Assessment and Mitigation Plan

Risk	Description	Impact	Mitigation Plan
Data Loss	Tasks may be lost due to system errors.	High	Add local data persistence and auto-save features.
User Confusion	UI can be confusing or difficult to navigate.	Medium	Follow minimalist design principles and perform usability checks.
Time Delay	Project development may not meet the deadline.	High	A strong commitment to the agreed project plan and milestones.
Technical Bugs	System errors or crashes.	Medium	Test and debug thoroughly before you submit your final version.
Scope Creep	Adding too many features.	Medium	Stick to defined scope.
Notification Failure	Alerts may not trigger correctly on all devices.	Medium	Test notifications across different devices; offer alternative visual cues in app.
Deadline Extension Misuse	Students may lose a sense of urgency if they overuse the extend feature.	Low	Show extension history on task details page. Highlight extended tasks in different style.

11. Architectural Pattern

The Study Planner Tool has a three-layer architecture pattern:

- Presentation Layer (UI).
- Business Logic Layer.
- Data Layer.

This pattern is chosen to keep the system simple, maintainable, and easy to extend.

12. Architectural Layers

Presentation Layer (UI)

Responsible for displaying all screens to the user and handling all interactions:

- **Dashboard Screen:** Summary of Tasks, Progress bar, focus mode button.
- **Focus Mode Screen:** Shows only the highest priority task with a countdown timer.
- **Task List Screen:** List of all tasks showing priority(color) and whether it is completed or not.
- **Add Task Form:** Form used to add a new task, includes every detail field.
- **Extend Deadline bottomSheet:** Bottom sheet that gives you quick options to extend task deadline by 1 day, 1 week, or 1 month.
- **Notifications Screen:** Screen that shows notifications for overdue tasks, and what you will be reminded of soon.
- **Stats & Progress Screen:** Weekly graphs and display of tasks completed.
- **Sort & Filter bottomSheet:** Bottom sheet that has the options to sort and filter your task list.

Data Layer

Responsible for storing and retrieving all task data locally:

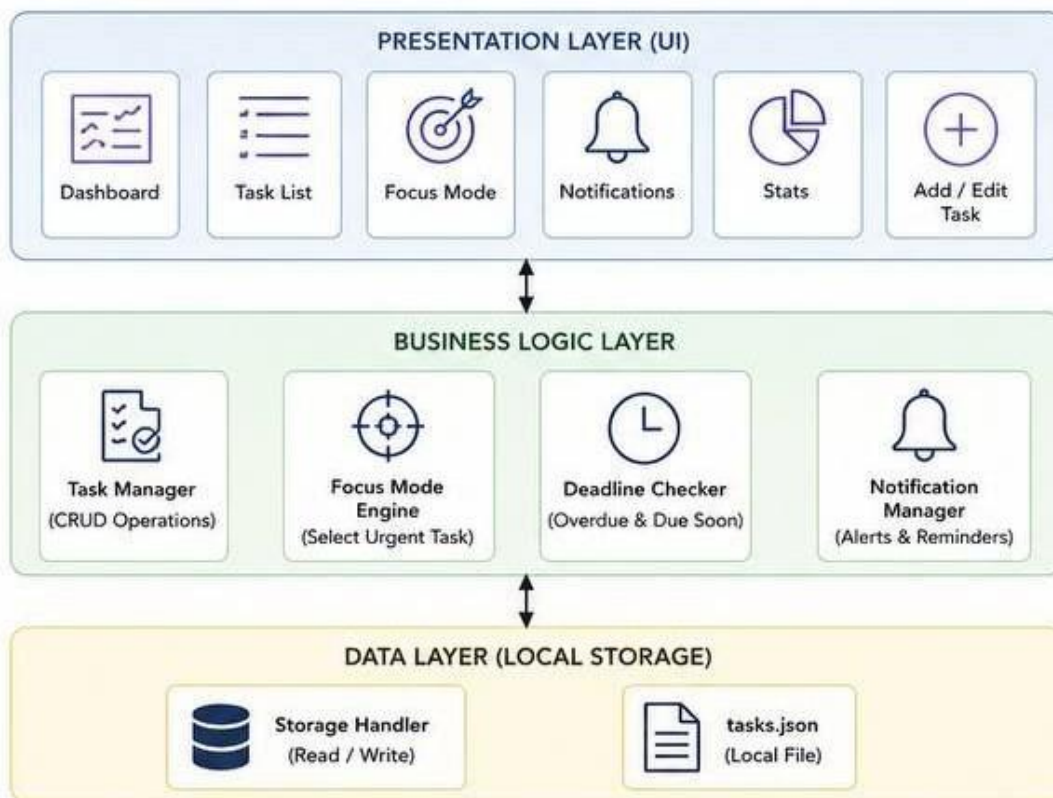
- **Storage Handler:** reads and writes task data to local storage.
- **tasks.json:** local file storing all task records.
- **Auto-save:** triggered automatically after every operation.

Business Logic Layer

Processes all user requests and applies system rules:

- **Task Manager:** Performs CRUD operations on tasks.
- **Focus Mode Engine:** Chooses the most prioritised task and minimizes everything else from the UI when Focus Mode is turned on.
- **Deadline Checker:** Identifies tasks that are overdue and due within the next 24 hours and signals color highlight function.
- **Extend Deadline Handler:** Processes requests to extend task deadlines, ensures the new date is valid, records deadline extension in history.
- **Notification Manager:** Alerts the user of overdue tasks and tasks that are due soon.
- **Filter & Sort Engine:** Sorts tasks by whatever the user chooses.
- **Validator:** Validates all fields before allowing the user to save a task.

13.Architectural Diagram



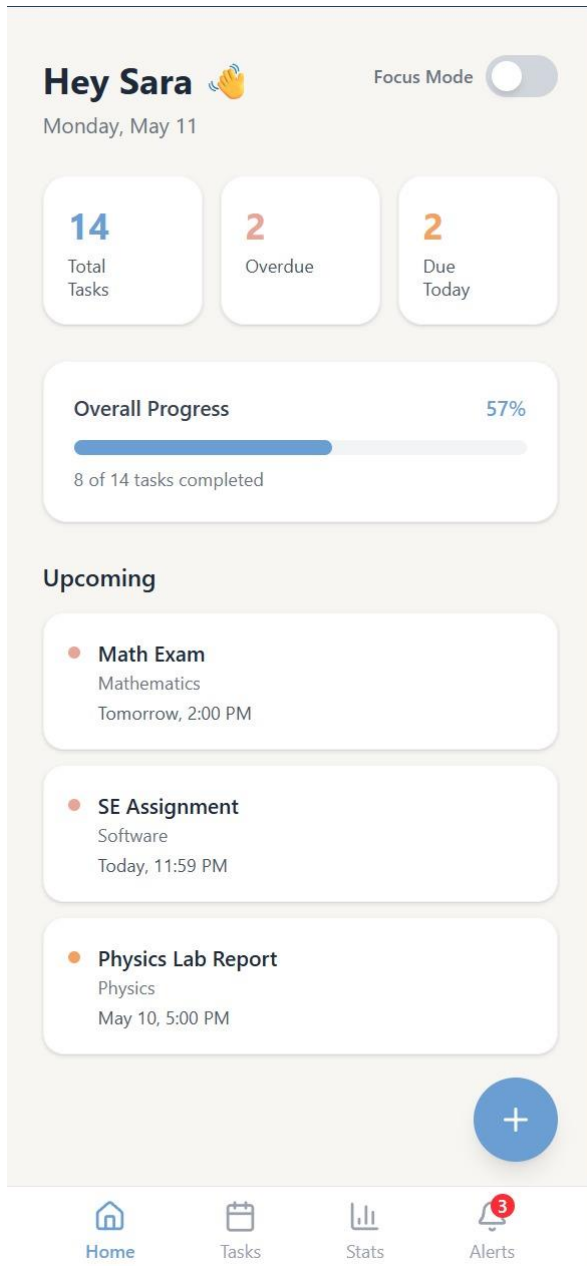
14. Component Responsibilities Table

Component	Layer	Responsibility
Dashboard Screen	Presentation	Show summary, progress, Focus Mode toggle.
Task List Screen	Presentation	Display all tasks color-coded by status.
Sort & Filter Panel	Presentation	Sort and filter tasks by criteria.
Add Task Form	Presentation	Create new task with all fields.
Extend Deadline Sheet	Presentation	Quick deadline extension UI.
Task Manager	Business Logic	CRUD operations on tasks.
Focus Mode Engine	Business Logic	Select urgent task, hide other elements.
Deadline Checker	Business Logic	Detect overdue and due-today tasks.
Extend Deadline Handler	Business Logic	Validate and save extended deadlines.
Notification Manager	Business Logic	Trigger overdue and reminder alerts.
Validator	Business Logic	Validate all inputs before saving.
Storage Handler	Data	Read and write tasks to local JSON.

15. UI Wireframes / Mockups

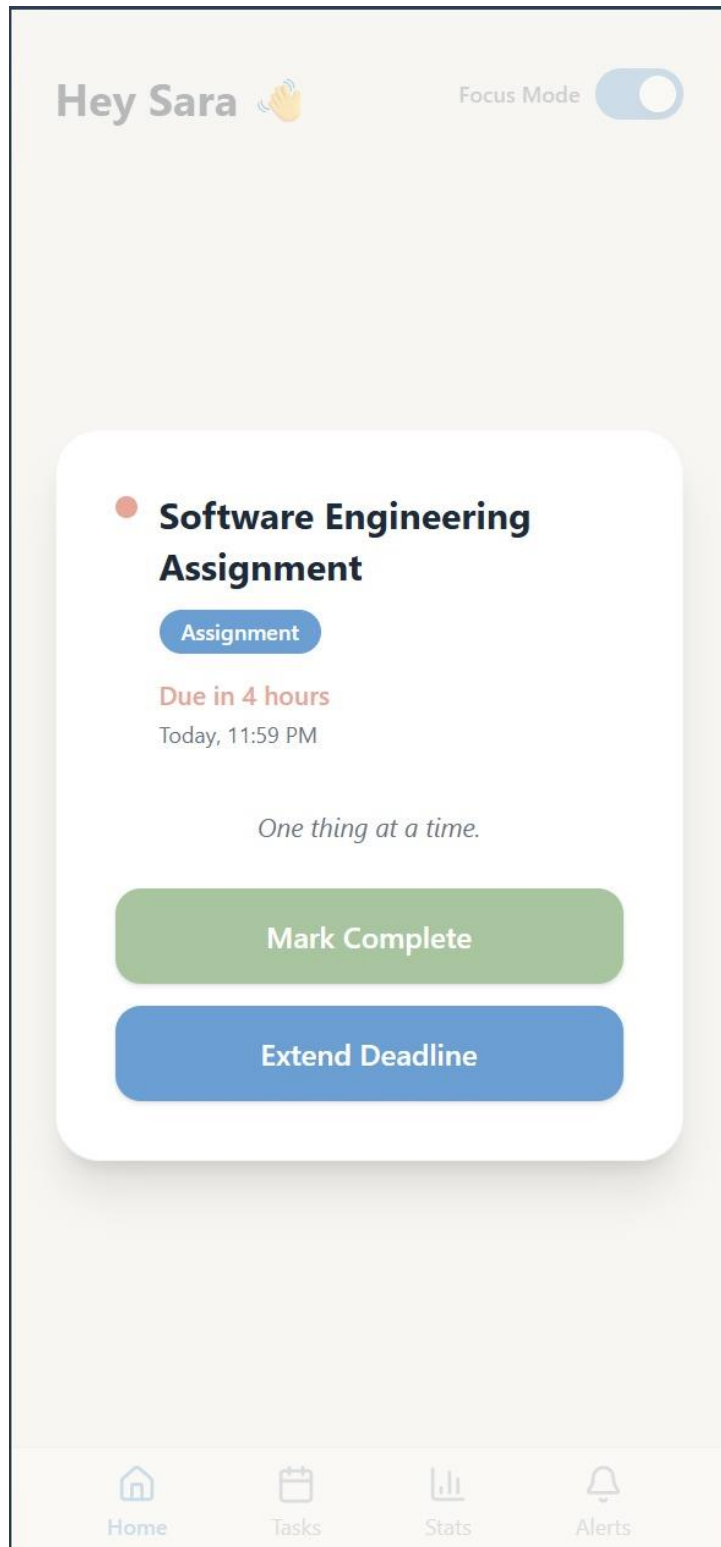
The following screens were designed using Figma. The designs cover the core functionalities of the system including task management, focus mode, filtering, notifications, and progress tracking.

Screen 1 - Dashboard (Normal Mode)



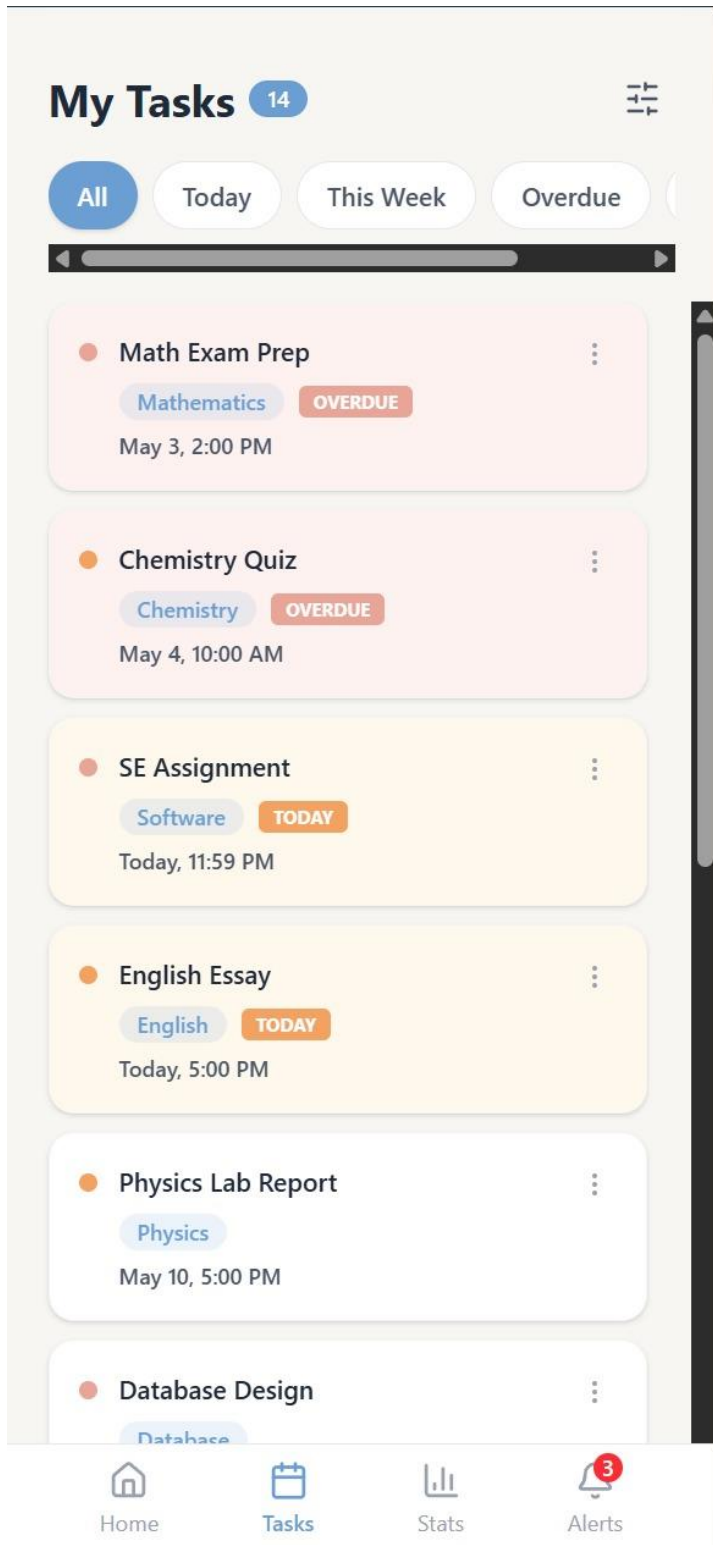
Displays task summary, progress bar, upcoming tasks, and Focus Mode toggle.

Screen 2 – Focus Mode



Displays the most urgent task with countdown and quick actions.

Screen 3 – Task List



Shows all tasks with sorting and filtering options.

Screen 4 – Add Task Form

 **New Task** 

What do you need to do?

SUBJECT / TYPE

 Assignment

 Exam

 Project

DEADLINE

Date

Time

dd/mm/yyyy

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PRIORITY

High

Medium

Low

ESTIMATED TIME

How long will this take?hours

NOTES (OPTIONAL)

Add any additional details...

Add Task

 Home

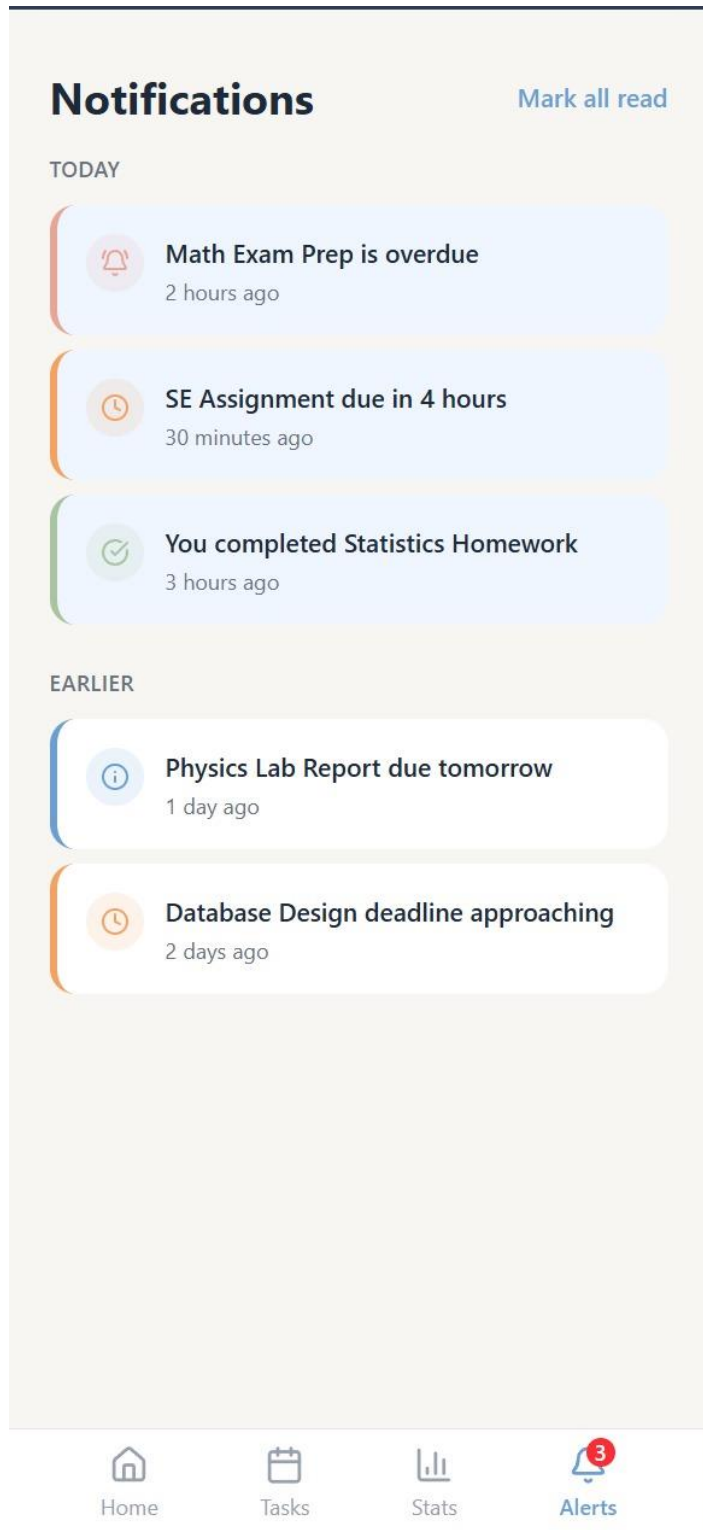
 Tasks

 Stats

 Alerts

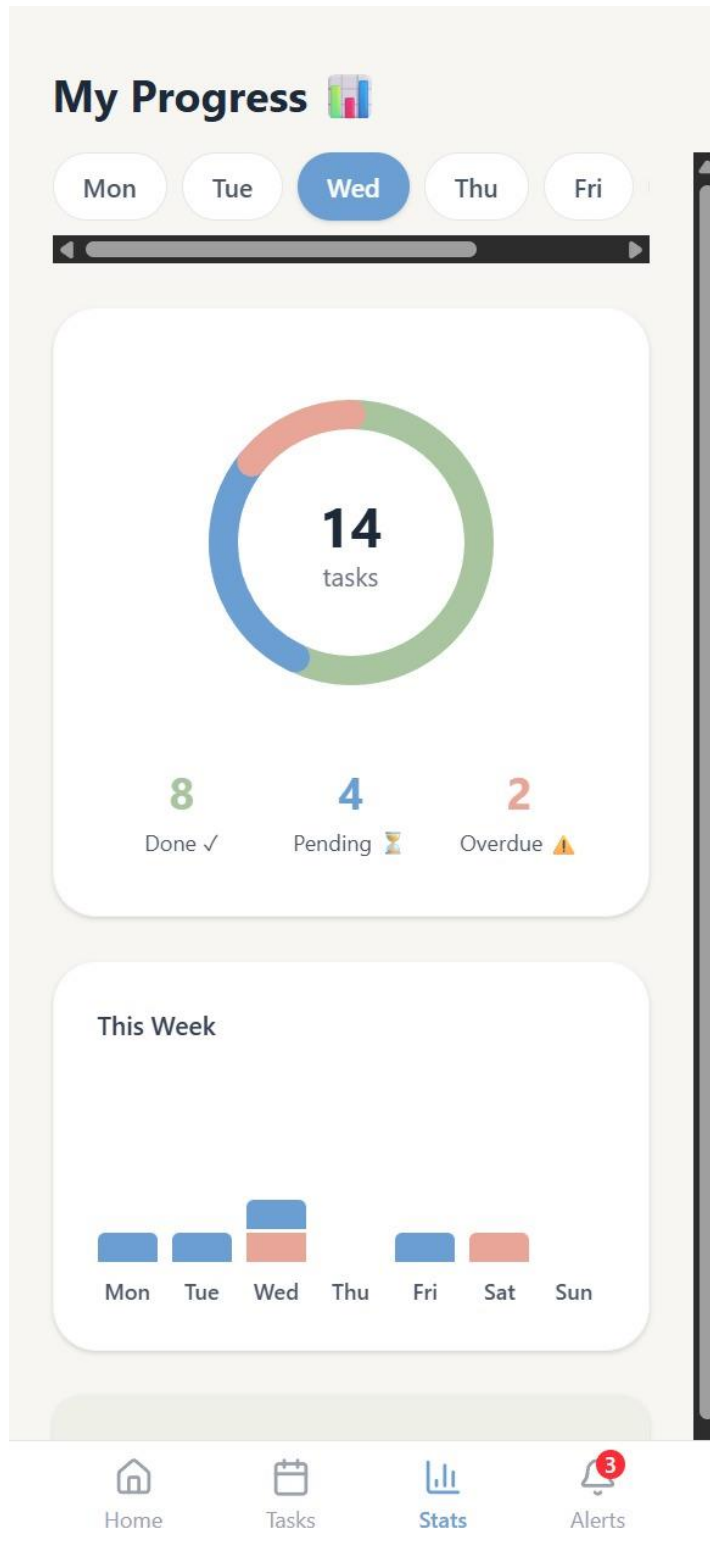
Allows the user to create a new task with task details.

Screen 5 – Notifications



Displays overdue, reminder, and completion notifications.

Screen 6 – Stats & Progress



Displays charts and weekly task statistics.

Screen 7 - Sort & Filter Panel



Provides sorting and filtering options for tasks.

16. UML Diagram with Explanation

Sequence Diagram

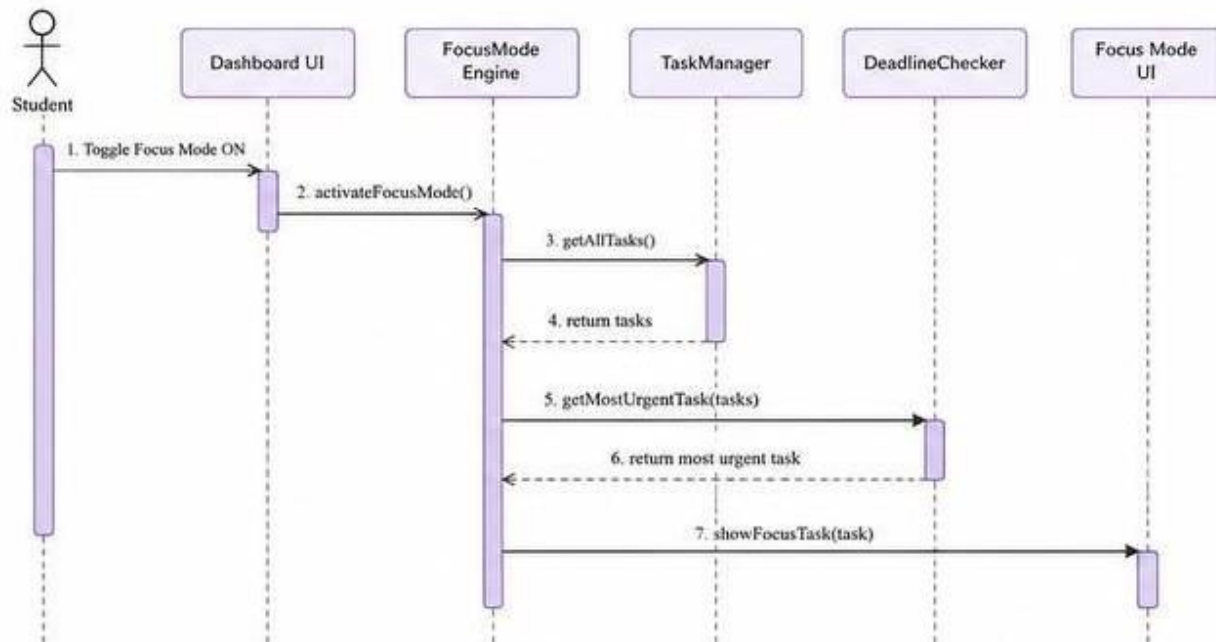
We used Sequence Diagram because this diagram shows the interaction between components as well as the order of which actions are performed by the user and it shows it very well.

Our diagram shows how the Study Planner Tool will carry out actions such as activating Focus Mode and extending deadlines one step at a time.

We did not use Class Diagram because our assignment shows more of the behavior of the system and how a user would interact with it rather than showing the details of the classes and objects. A sequence diagram fits our needs better.

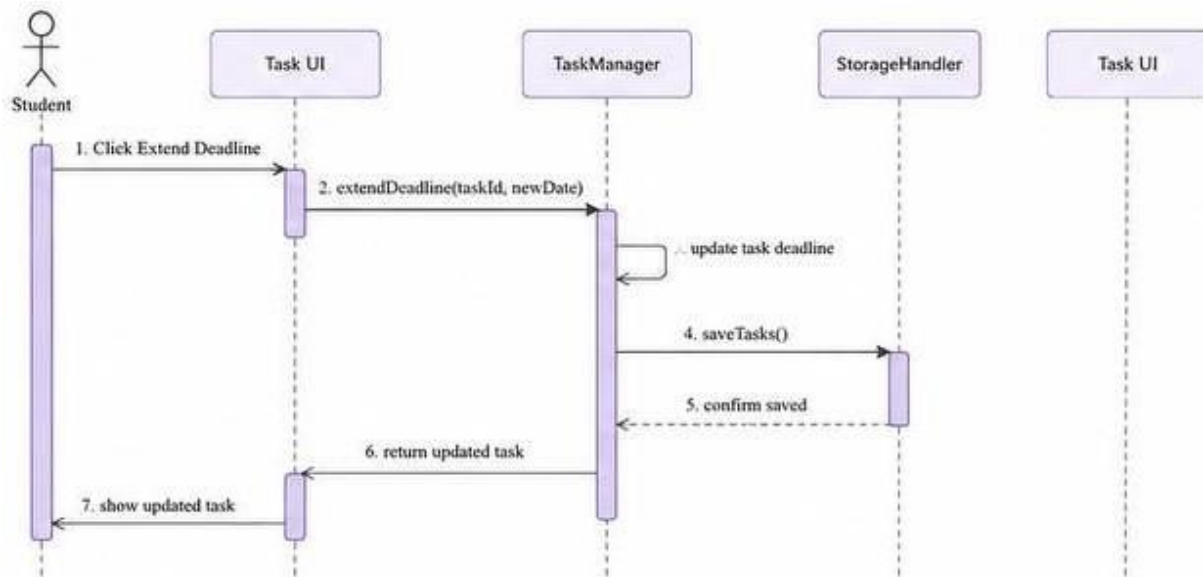
Sequence Diagram – Focus Mode Activation

1. User enables Focus Mode located on UI Dashboard.
2. UI Dashboard passes request to Focus Mode Engine.
3. Focus Mode Engine queries Task Manager for all active tasks.
4. Task Manager provides full list of tasks.
5. Focus Mode Engine passes list of tasks to Deadline Checker.
6. Deadline Checker relays the most urgent task
7. Focus Mode Engine passes urgent task back to UI.
8. UI hides everything else and displays Focus Mode screen with task card, countdown, and action buttons.



Sequence Diagram 2 – Extend Deadline

1. Student clicks Extend Deadline button on task card.
2. UI displays Extend Deadline bottom sheet.
3. Student selects extension option (+1 Day, +3 Days, +1 Week, or Custom).
4. Student inputs optional reason and clicks Confirm Extension.
5. UI posts new deadline info to Extend Deadline Handler.
6. Extend Deadline Handler requests Validator to ensure that the date is not in the past.
7. Validator responds with valid.
8. Extend Deadline Handler passes newly updated Task to Task Manager.
9. Task Manager requests Storage Handler to update task.
10. Storage Handler writes task to tasks.json
11. UI dismisses sheet and updates task card with new deadline. UI highlights task as extended.



17. Test Scenarios

Scenario 1 – Add a New Task Successfully

Description: The student opens the Add Task form, fills all required fields correctly, and submits the form. The system should save the task and display it in the Task List.

Preconditions:

- The application is open and running.
- The student is on the Dashboard or Task List screen.

Steps:

1. Student presses Add Task button.
2. Student types Title “Math Assignment”.
3. Student chooses type “Assignment”.
4. Student assigns future date and time due.
5. Student chooses priority “High”.
6. Student presses Add Task button to create task.

Expected Result: The task shows up in the Task List ordered by closest deadline and highlighted for priority and whether it is past due or not.

Scenario 2 – Activate Focus Mode

Description: Student turns on Focus Mode from the Dashboard. System should hide everything and only shows the most critical task.

Preconditions:

- The student is on the Dashboard.
- At least one task exists in the system.

Steps:

1. Student should see Dashboard in Normal Mode.
2. Student clicks Focus Mode toggle to ON.
3. System finds most urgent task.
4. Focus Mode screen appears.

Expected Result: Everything on the dashboard is hidden and only one task card should be shown with the most urgent task. It will show the task title, deadline countdown, priority badge, and two action buttons (Mark Complete and Extend Deadline).

Scenario 3 – Extend a Task Deadline

Description: The student extends the deadline of an existing task using the extension options. The system should reflect any changes in the Task List.

Preconditions:

- At least one task exists in the system.
- The student is viewing the task detail or Focus Mode.

Steps:

1. Student taps Extend Deadline button on a task.
2. Extend Deadline sheet appears.
3. Student selects “+3 Days” option.
4. Student enters optional reason “Need more time to study”.
5. Student taps Confirm Extension.

Expected Result: The deadline is updated by 3 days. The task card in the Task List shows the new deadline and is highlighted to indicate it has been extended. The sheet closes automatically.

18. Test Cases

Test Case 1 – Add Task with Valid Data

Test Case ID	TC-01.
Test Case Name	Add Task with Valid Data.
Related Scenario	Scenario 1.
Preconditions	App is open, student is on Dashboard.
Input Data	Title: "Math Assignment", Type: Assignment, Deadline: future date, Priority: High.
Steps	1. Tap Add Task button. 2. Fill all fields correctly. 3. Tap Add Task to submit.
Expected Result	Task saved successfully and appears in Task List sorted by deadline.
Actual Result	Task appears in Task List correctly.
Status	Pass.

Test Case 2 – Add Task with Empty Title

Test Case ID	TC-02.
Test Case Name	Add Task with Empty Title.
Related Scenario	Scenario 1.
Preconditions	App is open, student is on Add Task form.
Input Data	Title: empty, Type: Exam, Deadline: future date, Priority: Medium.
Steps	1. Open Add Task form. 2. Leave title field empty. 3. Tap Add Task to submit.
Expected Result	System displays validation error "Title cannot be empty" and does not save the task.
Actual Result	Validation error message displayed, task not saved.
Status	Pass.

Test Case 3 – Activate Focus Mode with Active Tasks

Test Case ID	TC-03.
Test Case Name	Activate Focus Mode with Active Tasks.
Related Scenario	Scenario 2.
Preconditions	App is open on Dashboard, at least one active task exists.
Input Data	Focus Mode toggle: ON.
Steps	1. View Dashboard in Normal Mode. 2. Tap Focus Mode toggle to ON.
Expected Result	All dashboard elements hidden. Most urgent task displayed with countdown, priority badge, Mark Complete and Extend Deadline buttons.
Actual Result	Focus Mode screen displayed correctly with most urgent task.
Status	Pass.

Test Case 4 – Activate Focus Mode with No Tasks

Test Case ID	TC-04.
Test Case Name	Activate Focus Mode with No Tasks.
Related Scenario	Scenario 2.
Preconditions	App is open on Dashboard, no tasks exist in the system.
Input Data	Focus Mode toggle: ON.
Steps	1. Ensure no tasks exist. 2. Tap Focus Mode toggle to ON.
Expected Result	System displays message "No tasks to focus on. Add a task first!" and toggle returns to OFF automatically.
Actual Result	Friendly message displayed, toggle returns to OFF.
Status	Pass.

Test Case 5 – Extend Deadline with Valid Option

Test Case ID	TC-05.
Test Case Name	Extend Deadline with Valid Option.
Related Scenario	Scenario 3.
Preconditions	At least one task exists, student is viewing task detail.
Input Data	Extension option: +3 Days, Reason: "Need more time".
Steps	1. Tap Extend Deadline. 2. Select +3 Days. 3. Enter reason. 4. Tap Confirm Extension.
Expected Result	Deadline updated by 3 days. Task card shows new deadline with extended highlight. Sheet closes automatically.
Actual Result	Deadline updated correctly, task card updated with new deadline.
Status	Pass.

Test Case 6 – Extend Deadline with Past Date

Test Case ID	TC-06.
Test Case Name	Extend Deadline with Past Date.
Related Scenario	Scenario 3.
Preconditions	At least one task exists, Extend Deadline sheet is open.
Input Data	Extension option: Custom, New Date: a date in the past.
Steps	1. Tap Extend Deadline. 2. Select Custom. 3. Enter a past date. 4. Tap Confirm Extension.
Expected Result	System displays validation error "Deadline must be a future date" and does not update the task.
Actual Result	Validation error displayed, deadline not changed.
Status	Pass.

19. Bug Report

Bug 1 – Focus Mode Toggle Does Not Reset When All Tasks Deleted

Bug ID	BUG-01.
Bug Title	Focus Mode stays ON after all tasks are deleted.
Severity	Medium.
Related Test Case	TC-04.
Steps to Reproduce	1. Add one task. 2. Activate Focus Mode. 3. Delete the task while Focus Mode is ON. 4. Return to Dashboard.
Expected Behavior	Focus Mode toggle should automatically switch OFF and show the empty state message.
Actual Behavior	Focus Mode toggle remains ON but the screen is blank with no content displayed.
Suggested Fix	Add a check in Focus Mode Engine: if task list is empty while Focus Mode is ON, automatically deactivate Focus Mode and display the empty state screen
Status	Identified — Not yet fixed

Bug 2 – Extend Deadline Sheet Does Not Close on Outside Tap

Bug ID	BUG-02
Bug Title	Extend Deadline sheet remains open when tapping outside
Severity	Low
Related Test Case	TC-05
Steps to Reproduce	5. Open any task. 6. Tap Extend Deadline. 7. Tap outside the sheet on the background.
Expected Behavior	Sheet should close and return to task detail screen
Actual Behavior	Sheet remains open and does not respond to outside tap
Suggested Fix	Add a dismiss on outside tap behavior to the Extend Deadline bottom sheet component
Status	Identified — Not yet fixed

20. References

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